«SpringBootApplication» Application
○ main(in args : String[]) : void
com
screenplay
security
«Component» JwtAuthenticationFilter
□ jwtUtil : JwtUtil
◇ doFilterInternal(in request : HttpServletRequest, inout response : HttpServletResponse, in chain : FilterChain) : void
□ extractJwtToken(in request : HttpServletRequest) : String
□ shouldProcessAuthentication(in username : String) : boolean
□ processAuthentication(in request : HttpServletRequest, in jwt : String, in username : String) : void
□ createUserDetailsFromToken(in jwt : String, in username : String) : UserDetails
□ setAuthenticationInContext(in request : HttpServletRequest, in userDetails : UserDetails) : void
calls
«Component» JwtUtil
□ secret : String
□ JWT_TOKEN_VALIDITY : long
□ getSigningKey() : SecretKey
□ getAllClaimsFromToken(in token : String) : Claims
○ getClaimFromToken(in token : String, in resolver : Function<Claims,T>) : T
○ getUsernameFromToken(in token : String) : String
○ getRoleFromToken(in token : String) : String
○ getExpirationDateFromToken(in token : String) : Date
□ isTokenExpired(in token : String) : boolean
○ generateToken(in username : String, in role : String) : String
□ doGenerateToken(in claims : Map<String,Object>, in subject : String) : String
○ validateToken(in token : String) : boolean
controller
«Controller» UserController
□ userService : UserService
○ getAllUsers(in page : int, in size : int, in search : String) : ResponseEntity<Page<User>>
○ deleteUser(in id : Long) : ResponseEntity<?>
○ toggleStatus(in id : Long) : ResponseEntity<User>
○ changeRole(in id : Long, in role : Role) : ResponseEntity<User>
«Controller» VideoController
□ videoService : VideoService
○ getPublishedVideos(in page : int, in size : int, in search : String) : ResponseEntity<Page<Video>>
○ getFeaturedVideos() : ResponseEntity<List<Video>>
○ getAdminVideos(in page : int, in size : int, in search : String) : ResponseEntity<Page<Video>>
○ getAdminStats() : ResponseEntity<VideoStatsResponse>
○ createVideo(in req : VideoRequest) : ResponseEntity<Video>
○ updateVideo(in id : Long, in req : VideoRequest) : ResponseEntity<Video>
○ deleteVideo(in id : Long) : ResponseEntity<?>
○ togglePublish(in id : Long, in value : boolean) : ResponseEntity<Video>
«Controller» AuthController
□ authService : AuthService
○ register(in req : RegisterRequest) : ResponseEntity<?>
○ login(in req : LoginRequest) : ResponseEntity<AuthResponse>
○ verifyEmail(in token : String) : ResponseEntity<?>
○ forgotPassword(in req : ForgotPasswordRequest) : ResponseEntity<?>
○ resetPassword(in req : ResetPasswordRequest) : ResponseEntity<?>
○ changePassword(in principal : Principal, in req : ChangePasswordRequest) : ResponseEntity<?>
«Controller» FileUploadController
□ fileService : FileService
○ uploadVideo(in file : MultipartFile) : ResponseEntity<?>
○ uploadImage(in file : MultipartFile) : ResponseEntity<?>
○ serveVideo(in uuid : String, in request : HttpServletRequest, inout response : HttpServletResponse) : void
○ serveImage(in uuid : String, inout response : HttpServletResponse) : void
«Controller» WatchlistController
□ watchlistService : WatchlistService
○ getWatchlist(in principal : Principal, in page : int, in size : int, in search : String) : ResponseEntity<Page<Video>>
○ addToWatchlist(in principal : Principal, in videoId : Long) : ResponseEntity<?>
○ removeFromWatchlist(in principal : Principal, in videoId : Long) : ResponseEntity<?>
configures
serviceImpl
«Service» FileServiceImpl
□ uploadDir : String
○ uploadVideo(in file : MultipartFile) : Map<String,String>
○ uploadImage(in file : MultipartFile) : Map<String,String>
○ serveVideo(in uuid : String, in request : HttpServletRequest, inout response : HttpServletResponse) : void
○ serveImage(in uuid : String, inout response : HttpServletResponse) : void
□ saveFile(in file : MultipartFile, in subDir : String) : String
«Service» AuthServiceImpl
□ userRepository : UserRepository
□ passwordEncoder : PasswordEncoder
□ jwtUtil : JwtUtil
□ emailService : EmailService
○ register(in req : RegisterRequest) : void
○ login(in req : LoginRequest) : AuthResponse
○ verifyEmail(in token : String) : void
○ forgotPassword(in req : ForgotPasswordRequest) : void
○ resetPassword(in req : ResetPasswordRequest) : void
○ changePassword(in email : String, in req : ChangePasswordRequest) : void
«Service» UserServiceImpl
□ userRepository : UserRepository
○ getAllUsers(in page : int, in size : int, in search : String) : Page<User>
○ deleteUser(in id : Long) : void
○ toggleUserStatus(in id : Long) : User
○ changeUserRole(in id : Long, in role : Role) : User
«Service» VideoServiceImpl
□ videoRepository : VideoRepository
□ userRepository : UserRepository
○ getPublishedVideos(in page : int, in size : int, in search : String) : Page<Video>
○ getFeaturedVideos() : List<Video>
○ getAdminVideos(in page : int, in size : int, in search : String) : Page<Video>
○ getAdminStats() : VideoStatsResponse
○ createVideo(in req : VideoRequest) : Video
○ updateVideo(in id : Long, in req : VideoRequest) : Video
○ deleteVideo(in id : Long) : void
○ togglePublish(in id : Long, in value : boolean) : Video
«Service» WatchlistServiceImpl
□ userRepository : UserRepository
□ videoRepository : VideoRepository
○ getWatchlist(in email : String, in page : int, in size : int, in search : String) : Page<Video>
○ addToWatchlist(in email : String, in videoId : Long) : void
○ removeFromWatchlist(in email : String, in videoId : Long) : void
service
«Interface» UserService
○ getAllUsers(in page : int, in size : int, in search : String) : Page<User>
○ deleteUser(in id : Long) : void
○ toggleUserStatus(in id : Long) : User
○ changeUserRole(in id : Long, in role : Role) : User
«Interface» FileService
○ uploadVideo(in file : MultipartFile) : Map<String,String>
○ uploadImage(in file : MultipartFile) : Map<String,String>
○ serveVideo(in uuid : String, in request : HttpServletRequest, inout response : HttpServletResponse) : void
○ serveImage(in uuid : String, inout response : HttpServletResponse) : void
«Interface» VideoService
○ getPublishedVideos(in page : int, in size : int, in search : String) : Page<Video>
○ getFeaturedVideos() : List<Video>
○ getAdminVideos(in page : int, in size : int, in search : String) : Page<Video>
○ getAdminStats() : VideoStatsResponse
○ createVideo(in req : VideoRequest) : Video
○ updateVideo(in id : Long, in req : VideoRequest) : Video
○ deleteVideo(in id : Long) : void
○ togglePublish(in id : Long, in value : boolean) : Video
«Interface» AuthService
○ register(in req : RegisterRequest) : void
○ login(in req : LoginRequest) : AuthResponse
○ verifyEmail(in token : String) : void
○ forgotPassword(in req : ForgotPasswordRequest) : void
○ resetPassword(in req : ResetPasswordRequest) : void
○ changePassword(in email : String, in req : ChangePasswordRequest) : void
«Interface» WatchlistService
○ getWatchlist(in email : String, in page : int, in size : int, in search : String) : Page<Video>
○ addToWatchlist(in email : String, in videoId : Long) : void
○ removeFromWatchlist(in email : String, in videoId : Long) : void
util
«Component» EmailService
□ mailSender : JavaMailSender
□ fromEmail : String
○ sendVerificationEmail(in toEmail : String, in token : String) : void
○ sendPasswordResetEmail(in toEmail : String, in token : String) : void
dao
«Repository» UserRepository
○ findByEmail(in email : String) : Optional<User>
○ existsByEmail(in email : String) : boolean
○ findByVerificationToken(in token : String) : Optional<User>
○ findByPasswordResetToken(in token : String) : Optional<User>
○ countByRoleAndActive(in role : Role, in active : boolean) : long
○ countByRole(in role : Role) : long
○ searchUsers(in search : String, in pageable : Pageable) : Page<User>
○ findWatchlistVideoIds(in email : String, in ids : List<Long>) : Set<Long>
○ findWatchlistByUserId(in userId : Long, in pageable : Pageable) : Page<Video>
○ searchWatchlistByUserId(in userId : Long, in search : String, in pageable : Pageable) : Page<Video>
«Repository» VideoRepository
○ searchVideos(in search : String, in pageable : Pageable) : Page<Video>
○ searchPublishedVideos(in search : String, in pageable : Pageable) : Page<Video>
○ findPublishedVideos(in pageable : Pageable) : Page<Video>
○ findRandomPublishedVideos(in pageable : Pageable) : List<Video>
○ countPublishedVideos() : long
○ getTotalDuration() : long
calls
realizes
extends
«use»
exception
«ControllerAdvice» GlobalExceptionHandler
○ handleResourceNotFound(in ex : ResourceNotFoundException) : ResponseEntity<?>
○ handleBadRequest(in ex : BadRequestException) : ResponseEntity<?>
○ handleUnauthorized(in ex : UnauthorizedException) : ResponseEntity<?>
○ handleGeneral(in ex : Exception) : ResponseEntity<?>
«handle»
ResourceNotFoundException
○ ResourceNotFoundException(in message : String)
BadRequestException
○ BadRequestException(in message : String)
UnauthorizedException
○ UnauthorizedException(in message : String)
entity
«Entity» User
□ id : Long
□ email : String
□ password : String
□ fullName : String
□ role : Role
□ active : boolean
□ emailVerified : boolean
□ verificationToken : String
□ verificationTokenExpiry : Instant
□ passwordResetToken : String
□ passwordResetTokenExpiry : Instant
□ createdAt : Instant
□ updatedAt : Instant
□ watchlist : Set<Video>
○ addToWatchlist(in video : Video) : void
○ removeFromWatchlist(in video : Video) : void
1
watchlist
0..*
«Entity» Video
□ id : Long
□ title : String
□ description : String
□ year : Integer
□ rating : String
□ duration : Integer
□ srcUuid : String
□ posterUuid : String
□ published : boolean
□ categories : List<String>
□ isInWatchlist : Boolean
□ createdAt : Instant
□ updatedAt : Instant
○ getSrc() : String
○ getPoster() : String
has
enums
«enumeration» Role
○ USER
○ ADMIN
config
«Configuration» SecurityConfig
□ jwtAuthFilter : JwtAuthenticationFilter
○ securityFilterChain(inout http : HttpSecurity) : SecurityFilterChain
○ passwordEncoder() : PasswordEncoder
○ authenticationManager(in config : AuthenticationConfiguration) : AuthenticationManager
«Configuration» CorsConfig
□ allowedOriginsRaw : String
○ addCorsMappings(inout registry : CorsRegistry) : void
creates & owns
SecurityFilterChain
WebMvcConfigurer
OncePerRequestFilter
RuntimeException
JpaRepository<User,Long>
JpaRepository<Video,Long>
dto
response
«DTO» UserStatsResponse
○ totalUsers : long
○ activeUsers : long
○ adminCount : long
«DTO» VideoStatsResponse
○ totalVideos : long
○ publishedVideos : long
○ totalDuration : long
«DTO» AuthResponse
○ token : String
○ role : String
○ email : String
○ fullName : String
request
«DTO» VideoRequest
○ title : String
○ description : String
○ year : Integer
○ rating : String
○ duration : Integer
○ src : String
○ poster : String
○ published : boolean
○ categories : List<String>
«DTO» RegisterRequest
○ fullName : String
○ email : String
○ password : String
«DTO» ChangePasswordRequest
○ currentPassword : String
○ newPassword : String
«DTO» ForgotPasswordRequest
○ email : String
«DTO» ResetPasswordRequest
○ token : String
○ newPassword : String
«DTO» LoginRequest
○ email : String
○ password : String

| Notation | Meaning |
|---|---|
| + | public |
| - | private |
| # | protected |
| ~ | package/default |
| in | input parameter |
| inout | bidirectional param |

| Arrow | Relationship |
|---|---|
| —▷ | Generalization |
| ··▷ | Realization |
| —> | Directed Assoc. |
| ◇— | Aggregation |
| ◆— | Composition |
| ··> | Dependency |
| ··> «use» | Usage |